

Jana Mahmoud

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EDUCATION

University of Toronto – Honours Bachelors of Science in Neuroscience (Minors: Psychology & Physiology) Sept 2022 – Apr 2026

Post-graduate Programming: Lab2Market Discover National Program – Aging and Brain Health Cohort May – July 2026

Selected participant in a national innovation program focused on translating research into scalable health solutions. Exploring challenges surrounding accessibility, implementation, and adoption of neuroimaging-based clinical technologies in diverse healthcare settings.

CORE QUALIFICATIONS

Education & Events: Speaker identification and outreach; seminars, rounds, webinars, workshops, and large-scale events; scheduling, registration, run-of-show, and attendance tracking; educational resource development; facilitator and stakeholder coordination

Funding & Partnerships: Sponsorship development; funding proposals and stewardship; budget coordination; relationship management across clinical, academic, nonprofit, and industry partners

Digital Communications: WordPress, Squarespace, Canva, Mailchimp, Microsoft Office, Google Workspace; website content, email campaigns, promotional posters, mailing lists, and digital archives; engagement and participation analysis

RELEVANT EXPERIENCE

Senior Thesis Researcher, Hodaie Lab, Toronto Western Hospital Krembil Brain Institute Sept 2025 – Apr 2026

Principal Investigator: Mojgan Hodaie, MD, MSc, FRCSC | Supervisor: Jerry Li, PhD Candidate | UofT Dept. of Surgery

Project: Pre-surgical MRI Morphometry and Clinical Variables Machine Learning-Based Prediction of Pain Recurrence Following Trigeminal Neuralgia (TN) Surgery (UofT Research Project in Human Biology, HMB496Y)

- Curated multimodal dataset of 350+ TN patients from TWH facial pain clinic, extracting clinical variables from EPIC EMR including demographics, pain characteristics, surgical history, and recurrence patterns.
- Developed machine learning classification models (XGBoost, SHAP) for individualized pain-free survival estimation and recurrence prediction; employing PyRadiomics for radiomic feature extraction from high-resolution MRI and implementing SHAP-based interpretability analyses to identify key predictive biomarkers.
- Translated findings for varied audiences through a national conference poster, an undergraduate research talk, lab presentations, and a manuscript in preparation. Contributed as a member of the Hodaie Lab with direct familiarity with the KBI research environment, Toronto Western Hospital operations, and the interdisciplinary teams supporting neuroscience care and discovery.

Incoming Co-President, Operations, NeuroTechUofT, Faculty of Applied Sciences & Engineering June 2026 – Present

- Selected to lead operations for the University of Toronto's only undergraduate neuroengineering design team, coordinating planning and accountability across neuroscience, software, hardware, education, community, and research portfolios.
- Manage funding and sponsorship strategy, budget coordination, and external relationships involving CRANIA, Krembil/UHN, SickKids, the Institute of Biomedical Engineering, and additional academic, nonprofit, and industry partners.
- Oversee member onboarding and training, workshops, speaker programming, observership opportunities, research showcases, and educational initiatives connecting students with clinicians, scientists, and engineers.
- Track executive and initiative progress through clear timelines, documentation, portfolio reporting, and cross-team communication while supporting clinically grounded neurotechnology projects.

Vice President of Events, UofT Pre-Medical Student Association (UTPMSA) Aug 2022 – Apr 2025

- Organized and executed major events including Club Fair, Program Fair, Temerty Faculty of Medicine Information Session, Mingle with Medical Students, Suturing Workshop, Study Seminar, and annual Medical Symposium serving 200-400+ attendees per event.
- Identified and invited speakers (professors and medical students); coordinated briefing, scheduling, registration, virtual and in-person logistics, audience questions, promotional materials, and post-event follow-up across three academic years.
- Created email and digital outreach campaigns, developed the UTPMSA Handbook as a lasting educational resource, secured more than \$10,000 in sponsorships, and maintained partner relationships.

Chief Executive Assistant, Office of the President, UofT Students' Union (UTSU) Jun 2024 – Apr 2025

- Coordinated daily operations, calendars, communications, and institutional documentation across a 25+ member executive body, triaging time-sensitive requests and maintaining continuity across overlapping portfolios.
- Authored 10+ research-driven white papers, interim reports, and briefing documents for senior stakeholders, including the Interim Food Insecurity Report examining student food access at the University of Toronto.
- Led recruitment, onboarding, and training for Executive Assistants and First Year Council; developed reusable materials, tracked assignments, and supported junior leaders through event planning and institutional processes.

DIGITAL COMMUNICATIONS & PROGRAM LEADERSHIP

Web Development Officer, promoted to Chief Marketing Officer, Federation of Canadian Secondary Students 2020 – 2022

- Directed digital outreach and content operations across social media, graphic design, video, web, and paid advertising, supervising an 11-person executive team and responding to public inquiries.
- Oversaw 90+ digital publications and social posts; coordinated website updates and content priorities through a weekly Microsoft Excel calendar that tracked deliverables, questions, feedback, and ownership.

Hackathon Director & Advisor, UTRAHacks, University of Toronto Robotics Association Jul 2023 – June 2024

- Directed a 12-person team delivering a 450+ participant event; recruited speakers and workshop facilitators, managed sponsors and university partners, and coordinated schedules, volunteers, judging, communications, and day-of operations.
- Secured more than \$17,000 in sponsorships and managed external media and promotional outreach with the Faculty of Applied Science & Engineering, U of T Robotics Institute, Major League Hacking, and industry partners.

Lead Peer Mentor, Trinity College Life Sciences First-Year Community, Faculty of Arts & Science Aug 2023 – Apr 2025

- Facilitated biweekly education sessions and one-on-one support for 30 first-year students across two cohorts; coordinated workshop content, attendance records, Quercus updates, and administrative follow-up with program staff.

Executive Advisor, UofT Undergraduate Research Student Association (URSA) Sept 2025 – Apr 2026

- Provided operational guidance across marketing, finance, mentorship, web development, and lab-tour portfolios; served as HR liaison and supported planning for the Tri-Campus Undergraduate Research Conference.

ADDITIONAL RESEARCH & HEALTHCARE SERVICE EXPERIENCE

Research Collaborator, Manning Lab, Hospital for Sick Children January 2026

Principal Investigator: Kathryn Manning, PhD | SickKids Neurosciences & Mental Health Program, UofT Dept. Medical Biophysics

Project: Pregnancy During the COVID-19 Pandemic (PdCP) Associated with Altered Infant Brain Connectivity: Evidence from a Multimodal Neurodevelopmental Pipeline (BrainHack Toronto '26)

- Developed a multimodal DTI and resting-state fMRI pipeline for an infant neurodevelopmental cohort; presented the structure-function framework at BrainHack Global Toronto 2026.

Thesis Researcher, Zhang Lab, St. Michael's Hospital, Unity Health Toronto Sept 2023 – Apr 2024

Principal Investigator: Haibo Zhang, MD | Supervisors: Cong Lu, MSc and Chris Wu, PhD Candidate | UofT Dept. of Anesthesiology

Project: Utilizing Dragonfly ORS Artificial Intelligence Tools for Improved Rabbit Lung Micro-CT Scan Image Segmentation in VILI/ARDS Analysis (UofT Research Opportunity Program, ROP299Y)

- Collaborated with an ICU pediatrician investigating ventilator-induced lung injury (VILI) and acute respiratory distress syndrome (ARDS); analyzed 100+ lung micro-CT scans and compared manual, machine-learning, and deep-learning segmentation approaches for VILI/ARDS-induced lung tissue.

Summer Research Fellow, Lefebvre Lab, Hospital for Sick Children May – Sept 2023

Principal Investigator: Julie Lefebvre, PhD | SickKids Neurosciences & Mental Health Program, UofT Dept. of Molecular Genetics

Medicine by Design Program, SSuRE Program (\$10,000 Studentship Award)

Project: ANGLER-FISH – An In Situ Approach to Resolve Related mRNAs in Neuronal Wiring and Cell Transplantation Models

- Collaborated on a Medicine by Design funded project with Leo Chou, PhD, developing cell replacement therapies for degenerative retina disease (retinitis pigmentosa) using spatial transcriptomic approaches.
- Created an immunohistochemistry mounting protocol for intact retina; implemented imaging pipeline using ImageJ to locate and quantify RNA species, assessing RNA transfer from human to host mouse cells.

Patient Visitor Volunteer, Toronto Western Hospital, University Health Network Feb 2026 – Present

- Support inpatients through patient-centred companionship, conversation and supervised walks; liaise with nursing and clinical staff to address patient and visitor concerns.

Head Swim Instructor & Lifeguard, UofT Hart House Fitness & Aquatics 2022 – 2024

- Planned and led swim lessons, delivered individualized instruction and maintained safe, supportive learning environments for adults, seniors, and children with varied abilities and confidence levels.

SELECTED RESEARCH COMMUNICATION

- **Mahmoud, J.**, Li, J., Garcia-Flores, E., Wolfensohn, A., Hodaie, M. Machine Learning-Based Prediction of Pain Recurrence Following Trigeminal Neuralgia Surgery. CAN-ACN 19th Canadian Neuroscience Meeting. May 2026. (Poster Presentation)
- **Mahmoud, J.**, Li, J. Machine Learning-Based Prediction of Pain Recurrence Following Trigeminal Neuralgia Surgery. Trinity College Undergraduate Research Conference (TCURC). Mar 2026. (Oral Presentation)
- **Mahmoud, J.** Neurological and Neurodegenerative Disease Student Association (NANSA) Research Seminar. Feb 2026. (Panelist)
- **Mahmoud, J.***, Bhatt, N.*, Vazquez, E., Batajoo, M., James, K. PDPC Functional Structural Multimodal Pipeline. BrainHack Global Toronto 2026. Jan 2026. (Oral Presentation)